



1. Description

1.1. Project

Project Name	cubemx
Board Name	custom
Generated with:	STM32CubeMX 6.18.0
Date	07/25/2026

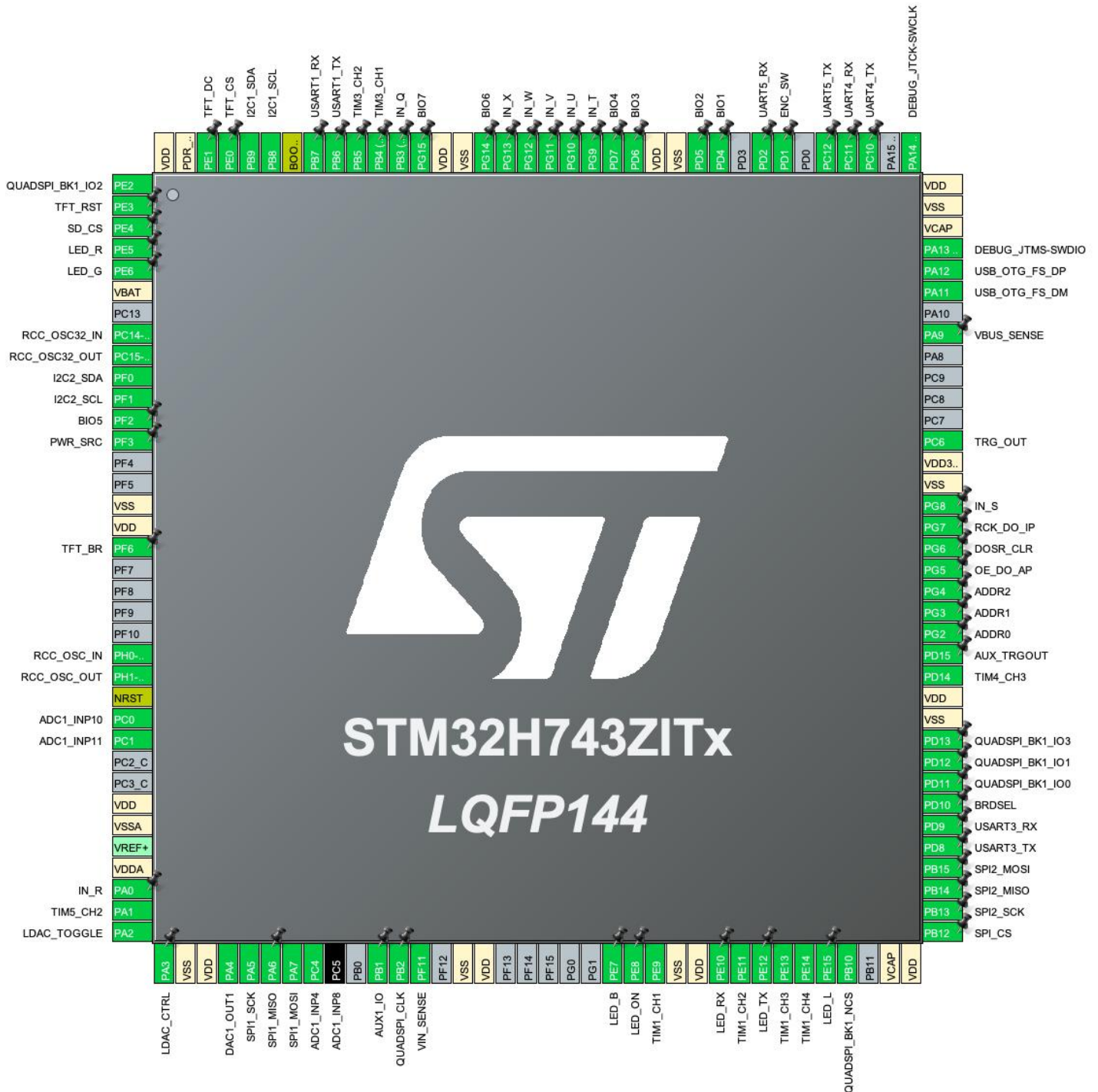
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H743/753
MCU name	STM32H743ZITx
MCU Package	LQFP144
MCU Pin number	144

1.3. Core(s) information

Core(s)	ARM Cortex-M7
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	QUADSPI_BK1_IO2	
2	PE3 *	I/O	GPIO_Output	TFT_RST
3	PE4 *	I/O	GPIO_Output	SD_CS
4	PE5 *	I/O	GPIO_Output	LED_R
5	PE6 *	I/O	GPIO_Output	LED_G
6	VBAT	Power		
8	PC14-OSC32_IN (OSC32_IN)	I/O	RCC_OSC32_IN	
9	PC15-OSC32_OUT (OSC32_OUT)	I/O	RCC_OSC32_OUT	
10	PF0	I/O	I2C2_SDA	
11	PF1	I/O	I2C2_SCL	
12	PF2 *	I/O	GPIO_Input	BIO5
13	PF3 *	I/O	GPIO_Input	PWR_SRC
16	VSS	Power		
17	VDD	Power		
18	PF6	I/O	TIM16_CH1	TFT_BR
23	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
24	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
25	NRST	Reset		
26	PC0	I/O	ADC1_INP10	
27	PC1	I/O	ADC1_INP11	
30	VDD	Power		
31	VSSA	Power		
33	VDDA	Power		
34	PA0	I/O	TIM2_ETR	IN_R
35	PA1	I/O	TIM5_CH2	
36	PA2	I/O	TIM2_CH3	LDAC_TOGGLE
37	PA3 *	I/O	GPIO_Output	LDAC_CTRL
38	VSS	Power		
39	VDD	Power		
40	PA4	I/O	DAC1_OUT1	
41	PA5	I/O	SPI1_SCK	
42	PA6	I/O	SPI1_MISO	
43	PA7	I/O	SPI1_MOSI	
44	PC4	I/O	ADC1_INP4	

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
45	PC5	I/O	ADC1_INP8	
47	PB1 *	I/O	GPIO_Input	AUX1_IO
48	PB2	I/O	QUADSPI_CLK	
49	PF11	I/O	ADC1_INP2	VIN_SENSE
51	VSS	Power		
52	VDD	Power		
58	PE7 *	I/O	GPIO_Output	LED_B
59	PE8 *	I/O	GPIO_Output	LED_ON
60	PE9	I/O	TIM1_CH1	
61	VSS	Power		
62	VDD	Power		
63	PE10 *	I/O	GPIO_Output	LED_RX
64	PE11	I/O	TIM1_CH2	
65	PE12 *	I/O	GPIO_Output	LED_TX
66	PE13	I/O	TIM1_CH3	
67	PE14	I/O	TIM1_CH4	
68	PE15 *	I/O	GPIO_Output	LED_L
69	PB10	I/O	QUADSPI_BK1_NCS	
71	VCAP	Power		
72	VDD	Power		
73	PB12 *	I/O	GPIO_Output	SPI_CS
74	PB13	I/O	SPI2_SCK	
75	PB14	I/O	SPI2_MISO	
76	PB15	I/O	SPI2_MOSI	
77	PD8	I/O	USART3_TX	
78	PD9	I/O	USART3_RX	
79	PD10 *	I/O	GPIO_Output	BRDSEL
80	PD11	I/O	QUADSPI_BK1_IO0	
81	PD12	I/O	QUADSPI_BK1_IO1	
82	PD13	I/O	QUADSPI_BK1_IO3	
83	VSS	Power		
84	VDD	Power		
85	PD14	I/O	TIM4_CH3	
86	PD15 *	I/O	GPIO_Output	AUX_TRGOUT
87	PG2 *	I/O	GPIO_Output	ADDR0
88	PG3 *	I/O	GPIO_Output	ADDR1
89	PG4 *	I/O	GPIO_Output	ADDR2
90	PG5 *	I/O	GPIO_Output	OE_DO_AP
91	PG6 *	I/O	GPIO_Output	DOSR_CLR

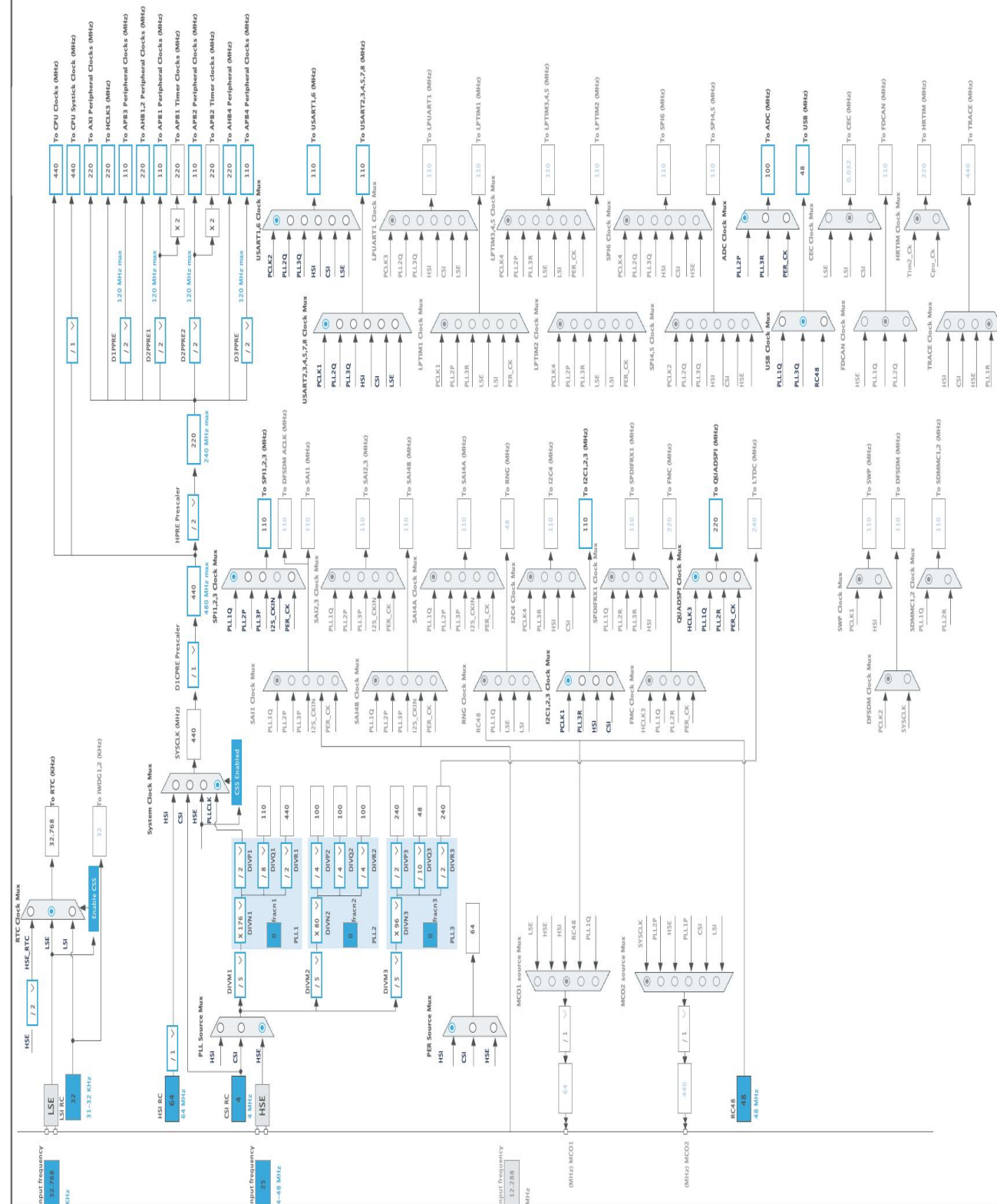
Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
92	PG7 *	I/O	GPIO_Output	RCK_DO_IP
93	PG8 *	I/O	GPIO_Input	IN_S
94	VSS	Power		
95	VDD33_USB	Power		
96	PC6	I/O	TIM8_CH1	TRG_OUT
101	PA9 *	I/O	GPIO_Input	VBUS_SENSE
103	PA11	I/O	USB_OTG_FS_DM	
104	PA12	I/O	USB_OTG_FS_DP	
105	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
106	VCAP	Power		
107	VSS	Power		
108	VDD	Power		
109	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
111	PC10	I/O	UART4_TX	
112	PC11	I/O	UART4_RX	
113	PC12	I/O	UART5_TX	
115	PD1 *	I/O	GPIO_Input	ENC_SW
116	PD2	I/O	UART5_RX	
118	PD4 *	I/O	GPIO_Input	BIO1
119	PD5 *	I/O	GPIO_Input	BIO2
120	VSS	Power		
121	VDD	Power		
122	PD6 *	I/O	GPIO_Input	BIO3
123	PD7 *	I/O	GPIO_Input	BIO4
124	PG9 *	I/O	GPIO_Input	IN_T
125	PG10 *	I/O	GPIO_Input	IN_U
126	PG11 *	I/O	GPIO_Input	IN_V
127	PG12 *	I/O	GPIO_Input	IN_W
128	PG13 *	I/O	GPIO_Input	IN_X
129	PG14 *	I/O	GPIO_Input	BIO6
130	VSS	Power		
131	VDD	Power		
132	PG15 *	I/O	GPIO_Input	BIO7
133	PB3 (JTDO/TRACESWO)	I/O	TIM2_CH2	IN_Q
134	PB4 (NJTRST)	I/O	TIM3_CH1	
135	PB5	I/O	TIM3_CH2	
136	PB6	I/O	USART1_TX	
137	PB7	I/O	USART1_RX	
138	BOOT0	Boot		

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
139	PB8	I/O	I2C1_SCL	
140	PB9	I/O	I2C1_SDA	
141	PE0 *	I/O	GPIO_Output	TFT_CS
142	PE1 *	I/O	GPIO_Output	TFT_DC
143	PDR_ON	Power		
144	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32H743ZITx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H743/753
MCU	STM32H743ZITx
Datasheet	DS12110 Rev8

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

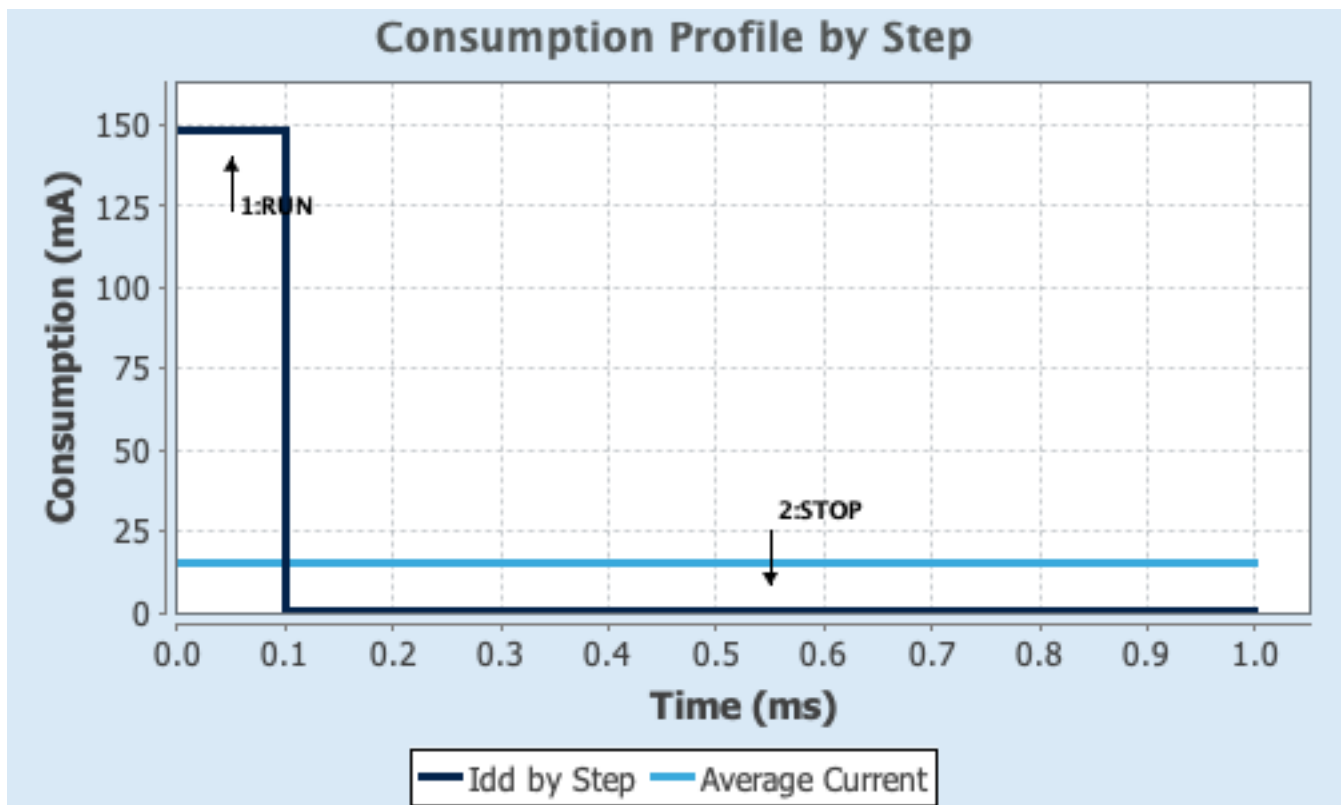
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0-High	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	ITCM	NA
CPU Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL	Flash-OFF
Clock Source Frequency	24 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	148 mA	150 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Ta Max	105.46	124.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	14.94 mA
Battery Life	1 day, 17 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	cubemx
Project Folder	/Users/gordonanderson/GAACE/Projects/MIPS_STM32/cubemx
Toolchain / IDE	Makefile
Firmware Package Name and Version	STM32Cube FW_H7 V1.13.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	No
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_ADC1_Init	ADC1
5	MX_DAC1_Init	DAC1
6	MX_I2C1_Init	I2C1
7	MX_I2C2_Init	I2C2
8	MX_QUADSPI_Init	QUADSPI
9	MX_SPI1_Init	SPI1
10	MX_SPI2_Init	SPI2
11	MX_TIM1_Init	TIM1

Rank	Function Name	Peripheral Instance Name
12	MX_TIM2_Init	TIM2
13	MX_TIM3_Init	TIM3
14	MX_TIM4_Init	TIM4
15	MX_TIM5_Init	TIM5
16	MX_TIM7_Init	TIM7
17	MX_TIM8_Init	TIM8
18	MX_UART4_Init	UART4
19	MX_UART5_Init	UART5
20	MX_USART1_UART_Init	USART1
21	MX_USART3_UART_Init	USART3
22	MX_USB_OTG_FS_PCD_Init	USB_OTG_FS
23	MX_RTC_Init	RTC
24	MX_TIM16_Init	TIM16

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN2: IN2 Single-ended

IN4: IN4 Single-ended

mode: IN8

IN10: IN10 Single-ended

mode: IN11

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Asynchronous clock mode divided by 1

Resolution ADC 16-bit resolution

Scan Conversion Mode Enabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

End Of Conversion Selection End of single conversion

Overrun behaviour Overrun data preserved

Left Bit Shift No bit shift

Conversion Data Management Mode DMA One Shot Mode

Low Power Auto Wait Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Oversampling Ratio 1

Number Of Conversion **5 ***

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel **Channel 10 ***

Sampling Time 1.5 Cycles

Offset Number No offset

Offset Signed Saturation Disable

Rank **2 ***

Channel **Channel 11 ***

Sampling Time 1.5 Cycles

Offset Number No offset

Offset Signed Saturation Disable

<u>Rank</u>	3 *
Channel	Channel 4 *
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset Signed Saturation	Disable
<u>Rank</u>	4 *
Channel	Channel 8 *
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset Signed Saturation	Disable
<u>Rank</u>	5 *
Channel	Channel 2
Sampling Time	64.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable
ADC_Injected_ConversionMode:	
Enable Injected Conversions	Disable
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	
Enable Analog WatchDog3 Mode	false

3.2. CORTEX_M7

3.2.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode	Enabled *
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Cortex Interface Settings:

CPU ICache	Disabled
CPU DCache	Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers
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Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled
MPU Region Base Address	0x0 *

MPU Region Size	4GB
MPU SubRegion Disable	0x87 *
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	DISABLE
MPU Shareability Permission	ENABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 2 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 13 Settings:

MPU Region	Disabled
------------	----------

Cortex Memory Protection Unit Region 14 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 15 Settings:

MPU Region	Disabled
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3.3. DAC1

OUT1 connected to: only external pin

3.3.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
Trigger	None
User Trimming	Factory trimming

3.4. DEBUG

Debug: Serial Wire

3.5. I2C1

I2C: I2C

3.5.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x40505681 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.6. I2C2

I2C: I2C

3.6.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x40505681 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.7. MEMORYMAP

mode: Activated

3.8. QUADSPI

QuadSPI Mode: Bank1 with Quad SPI Lines

3.8.1. Parameter Settings:

General Parameters:

Clock Prescaler	255
Fifo Threshold	1
Sample Shifting	No Sample Shifting
Flash Size	1
Chip Select High Time	1 Cycle
Clock Mode	Low
Flash ID	Flash ID 1
Dual Flash	Disabled

3.9. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.9.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
LSE Drive Capability	LSE oscillator low drive capability
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	3 WS (4 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 4 and 8 MHz
PLL2 input frequency range	Between 4 and 8 MHz
PLL3 input frequency range	Between 4 and 8 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	Wide VCO range
PLL3 clock Output range	Wide VCO range

3.10. RTC

mode: Activate Clock Source

mode: Activate Calendar

3.10.1. Parameter Settings:

General:

Hour Format	Hourformat 24
Asynchronous Predivider value	127
Synchronous Predivider value	255

Calendar Time:

Data Format	BCD data format
Hours	0
Minutes	0
Seconds	0
Day Light Saving: value of hour adjustment	Daylightsaving None
Store Operation	Storeoperation Reset

Calendar Date:

Week Day	Monday
Month	January
Date	1
Year	0

3.11. SPI1

Mode: Full-Duplex Master

3.11.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	32 *
Baud Rate	3.4375 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.12. SPI2

Mode: Full-Duplex Master

3.12.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	32 *
Baud Rate	3.4375 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.13. SYS

Timebase Source: SysTick

3.14. TIM1

Clock Source : Internal Clock

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2

Channel3: PWM Generation CH3

Channel4: PWM Generation CH4

3.14.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
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Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.15. TIM2

Slave Mode: Reset Mode

Trigger Source: ETR1

Clock Source : Internal Clock

Channel2: Input Capture direct mode

Channel3: Output Compare CH3

3.15.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295

Internal Clock Division (CKD)	No Division
auto-reload preload	Disable
Slave Mode Controller	Reset Mode
Trigger Output (TRGO) Parameters:	
Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Clear Input:	
Clear Input Source	Disable
Trigger:	
Trigger Polarity	non inverted
Trigger Prescaler	Prescaler not used
Trigger Filter (4 bits value)	0
Input Capture Channel 2:	
Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0
Output Compare Channel 3:	
Mode	Frozen (used for Timing base)
Pulse (32 bits value)	0
Output compare preload	Disable
CH Polarity	High

3.16. TIM3

Combined Channels: Encoder Mode

3.16.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
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____ Parameters for Channel 1 ____

Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

____ Parameters for Channel 2 ____

Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.17. TIM4

Clock Source : Internal Clock

Channel3: Output Compare CH3

3.17.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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Output Compare Channel 3:

Mode	Frozen (used for Timing base)
Pulse (16 bits value)	0
Output compare preload	Disable
CH Polarity	High

3.18. TIM5

Clock Source : Internal Clock

Channel2: Output Compare CH2

3.18.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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Output Compare Channel 2:

Mode	Frozen (used for Timing base)
Pulse (32 bits value)	0
Output compare preload	Disable
CH Polarity	High

3.19. TIM7

mode: Activated3.19.1. Parameter Settings:**Counter Settings:**

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)
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3.20. TIM8

Clock Source : Internal Clock**Channel1: Output Compare CH1**3.20.1. Parameter Settings:**Counter Settings:**

Prescaler (PSC - 16 bits value)	0
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Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
--------------------	---------

Output Compare Channel 1:

Mode	Frozen (used for Timing base)
Pulse (16 bits value)	0
Output compare preload	Disable
CH Polarity	High
CH Idle State	Reset

3.21. TIM16

mode: Activated

Channel1: PWM Generation CH1

3.21.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.22. UART4

Mode: Asynchronous

3.22.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.23. UART5

Mode: Asynchronous

3.23.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1

Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.24. USART1

Mode: Asynchronous

3.24.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.25. USART3

Mode: Asynchronous

3.25.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.26. USB_OTG_FS

Mode: Device_Only

3.26.1. Parameter Settings:

Speed	Full Speed 12MBit/s
Enable internal IP DMA	Disabled
Low power	Disabled
Battery charging	Disabled
Link Power Management	Disabled
Use dedicated end point 1 interrupt	Disabled

VBUS sensing	Disabled
Signal start of frame	Disabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC0	ADC1_INP10	Analog mode	No pull-up and no pull-down	n/a	
	PC1	ADC1_INP11	Analog mode	No pull-up and no pull-down	n/a	
	PC4	ADC1_INP4	Analog mode	No pull-up and no pull-down	n/a	
	PC5	ADC1_INP8	Analog mode	No pull-up and no pull-down	n/a	
	PF11	ADC1_INP2	Analog mode	No pull-up and no pull-down	n/a	VIN_SENSE
DAC1	PA4	DAC1_OUT1	Analog mode	No pull-up and no pull-down	n/a	
DEBUG	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
I2C1	PB8	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB9	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
I2C2	PF0	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PF1	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
QUADSPI	PE2	QUADSPI_BK1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB2	QUADSPI_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB10	QUADSPI_BK1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD11	QUADSPI_BK1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD12	QUADSPI_BK1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD13	QUADSPI_BK1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
RCC	PC14-OSC32_IN (OSC32_IN)	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SPI2	PB13	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM1	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE11	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE13	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE14	TIM1_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM2	PA0	TIM2_ETR	Alternate Function Push Pull	No pull-up and no pull-down	Low	IN_R
	PA2	TIM2_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	LDAC_TOGGLE
	PB3 (JTDO/TRACESWO)	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	IN_Q
TIM3	PB4 (NJTRST)	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB5	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM4	PD14	TIM4_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM5	PA1	TIM5_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM8	PC6	TIM8_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	TRG_OUT
TIM16	PF6	TIM16_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	TFT_BR
UART4	PC10	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC11	UART4_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UART5	PC12	UART5_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD2	UART5_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART1	PB6	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB7	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART3	PD8	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD9	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USB_OTG_FS	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PE3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TFT_RST
	PE4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SD_CS
	PE5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_R
	PE6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_G
	PF2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BIO5

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PF3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	PWR_SRC
	PA3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LDAC_CTRL
	PB1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	AUX1_IO
	PE7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_B
	PE8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_ON
	PE10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_RX
	PE12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_TX
	PE15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_L
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI_CS
	PD10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BRDSEL
	PD15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	AUX_TRGOUT
	PG2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ADDR0
	PG3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ADDR1
	PG4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ADDR2
	PG5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OE_DO_AP
	PG6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DOSR_CLR
	PG7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RCK_DO_IP
	PG8	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IN_S
	PA9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	VBUS_SENSE
	PD1	GPIO_Input	Input mode	Pull-up *	n/a	ENC_SW
	PD4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BIO1
	PD5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BIO2
	PD6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BIO3
	PD7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BIO4
	PG9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IN_T
	PG10	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IN_U
	PG11	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IN_V
	PG12	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IN_W
	PG13	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	IN_X
	PG14	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BIO6
	PG15	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BIO7
	PE0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TFT_CS
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TFT_DC

4.2. DMA configuration

DMA request	Stream	Direction	Priority
ADC1	DMA1_Stream0	Peripheral To Memory	Low

ADC1: DMA1_Stream0 DMA request Settings:

Mode: **Circular ***
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Half Word
Memory Data Width: Half Word

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

nothing configured in DMA service

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
DMA1 stream0 global interrupt	true	0	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupts	unused		
TIM1 break interrupt	unused		
TIM1 update interrupt	unused		
TIM1 trigger and commutation interrupts	unused		
TIM1 capture compare interrupt	unused		
TIM2 global interrupt	unused		
TIM3 global interrupt	unused		
TIM4 global interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
I2C2 event interrupt	unused		
I2C2 error interrupt	unused		
SPI1 global interrupt	unused		
SPI2 global interrupt	unused		
USART1 global interrupt	unused		
USART3 global interrupt	unused		
TIM8 break interrupt and TIM12 global interrupt	unused		
TIM8 update interrupt and TIM13 global interrupt	unused		
TIM8 trigger and commutation interrupts and TIM14 global interrupt	unused		
TIM8 capture compare interrupt	unused		
TIM5 global interrupt	unused		
UART4 global interrupt	unused		
UART5 global interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts		unused	
TIM7 global interrupt		unused	
FPU global interrupt		unused	
QUADSPI global interrupt		unused	
USB On The Go FS End Point 1 Out global interrupt		unused	
USB On The Go FS End Point 1 In global interrupt		unused	
USB OTG FS global interrupt		unused	
TIM16 global interrupt		unused	
HSEM1 global interrupt		unused	

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	true
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
DMA1 stream0 global interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Category view Power Domain view

 Choose filters by Power Domain

☐ D1 ☐ D2 ☐ D3 ☒ None

Middleware

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Other
BDMA	ADC1 ✓	RTC ✓	I2C1 ✓				DEBUG ✓		
CORTEX_M7 ✓	DAC1 ✓	TIM1 ✓	I2C2 ✓						
DMA ✓		TIM2 ✓	QUADSPI ✓						
GPIO ✓		TIM3 ✓	SPI1 ✓						
MDMA		TIM4 ✓	SPI2 ✓						
NVIC ✓		TIM5 ✓	UART4 ✓						
RCC ✓		TIM7 ✓	UART5 ✓						
SYS ✓		TIM8 ✓	USART1 ✓						
		TIM16 ✓	USART3 ✓						
			USB_FS ✓						

5.1.2. Without filters

Category view Power Domain view

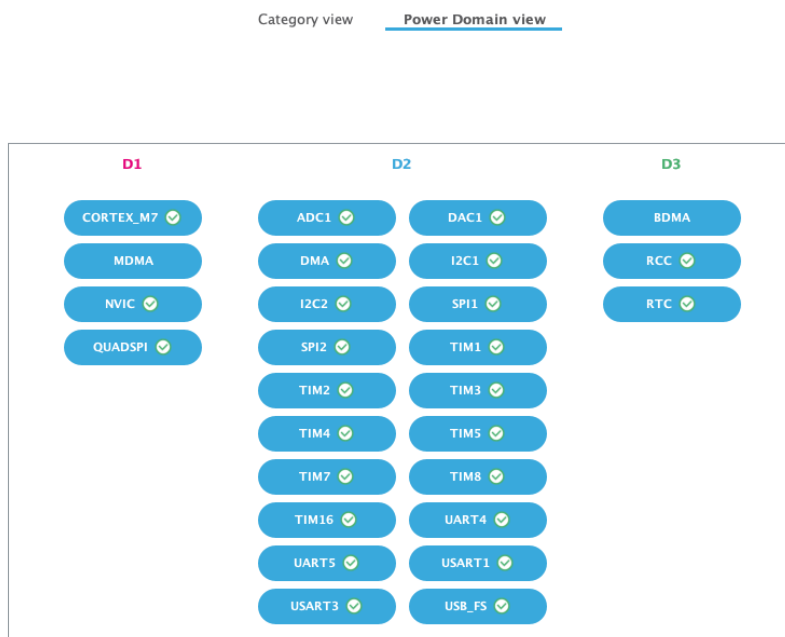
 Choose filters by Power Domain

☐ D1 ☐ D2 ☐ D3 ☒ None

Middleware

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Other
BDMA	ADC1 ✓	RTC ✓	I2C1 ✓				DEBUG ✓		
CORTEX_M7 ✓	DAC1 ✓	TIM1 ✓	I2C2 ✓						
DMA ✓		TIM2 ✓	QUADSPI ✓						
GPIO ✓		TIM3 ✓	SPI1 ✓						
MDMA		TIM4 ✓	SPI2 ✓						
NVIC ✓		TIM5 ✓	UART4 ✓						
RCC ✓		TIM7 ✓	UART5 ✓						
SYS ✓		TIM8 ✓	USART1 ✓						
		TIM16 ✓	USART3 ✓						
			USB_FS ✓						

5.2. Power Domain view



6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Brochures	https://www.st.com/resource/en/brochure/humanoid-robot-reference-guide.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-

protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5312-migration-from-revy-to-revv-for-stm32h743753-and-stm32h750-value-line-microcontrollers-stmicroelectronics.pdf

- Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4938-getting-started-with-stm32h74xig-and-stm32h75xig-mcu-hardware-development-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4230-introduction-to

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